Patuakhali Science and Technology University
Faculty of Computer Science and Engineering

CCE 310 :: Software Development Project-I

Project Report

Project Title : Reuse Hub
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Reuse Hub

1. Introduction

Reuse Hub is a cross-platform application designed to promote sustainable living through community-driven item reuse. It connects donors and seekers, enabling free exchange of usable goods with an intuitive and secure interface. Built using Flutter and powered by Supabase, the app ensures real-time interactions and platform independence. Key features include item listings, request handling, messaging, and role-based access. Reuse Hub aims to reduce waste while fostering a culture of sharing and environmental responsibility.

2. Objectives

- To develop a unified, cross-platform platform for donating and requesting reusable household items.
- Implement secure login and user roles.
- Enable item listing, browsing, and filtering.
- Support direct requests and real-time messaging.
- Ensure responsive design across mobile and web platforms.

3. Problem Statement

Existing item reuse and donation platforms are often fragmented, platform-dependent, or limited in functionality. Many rely on outdated interfaces, lack mobile responsiveness, or operate through third-party social groups with no automation. Key features such as real-time messaging, role-based access, and location-aware discovery are often missing. There is a clear need for a modern, cross-platform solution that streamlines item reuse while promoting sustainability and user convenience.

4. Related Work

- **OLIO** - A widely-used food and item sharing app. However, it is heavily localized and lacks full-featured support for non-food items.[1]
- **Freecycle** - A nonprofit global network for item reuse, but suffers from outdated UI, limited mobile experience, and lacks in-app messaging.
- **Letgo (Now OLX)** - Supported local item sharing. Shifted focus to resale, reducing emphasis on community reuse.[2]

- **Trash Nothing** - Focused purely on free giving, but lacks a modern UI and role-based access.
- **Facebook Marketplace** - Highly popular but lacks filters specifically for "free" items.
- **Buy Nothing Project** - A community-driven model restricted to Facebook Groups, lacking automated request handling and notifications.
- **Too Good to Go** - Focuses only on surplus food donations, with no support for general household items.

5. Scope

Reuse Hub is a cross-platform application built using Flutter, designed to facilitate the reuse and redistribution of pre-owned items within communities. The platform will enable users to donate or request items through a streamlined, user-friendly interface. Core functionalities include user authentication, item listings, request handling, messaging, and category-based browsing. The application ensures accessibility across Android, iOS, and web platforms.

6. Methodology

6.1. Technology Stack

The development of Reuse Hub will follow an agile methodology, allowing for iterative improvements and user feedback. Our technology stack will include,

- **Frontend**: Flutter (Dart)[3]
- **Backend**: Supabase (or Firebase)[4]
- **UI/ UX Design**: Figma
- **Database**: PostgreSQL
- **Authentication**: JWT or OAuth2
- **Hosting (web app)**: Vercel or Netlify
- **CI/CD**: GitHub Actions, Docker (optional)

6.2. Design Principles

The design of Reuse Hub will adhere to the following principles,

- **Material Design**: Adopts Google's Material Design guidelines to ensure a consistent and modern UI/UX across Android devices.
- **Responsive Design**: Guarantees seamless performance across various screen sizes and orientations, including desktop, tablet, and mobile views.
- **User-Centric Design**: Prioritizes user experience with intuitive navigation, minimal steps, and clear visual elements for both donors and receivers.
- **Documentation**: Provides thorough documentation for developers (e.g., API docs) and end-users (e.g., help guides, FAQs).

7. Visual Models

7.1. Flow Chart Diagram

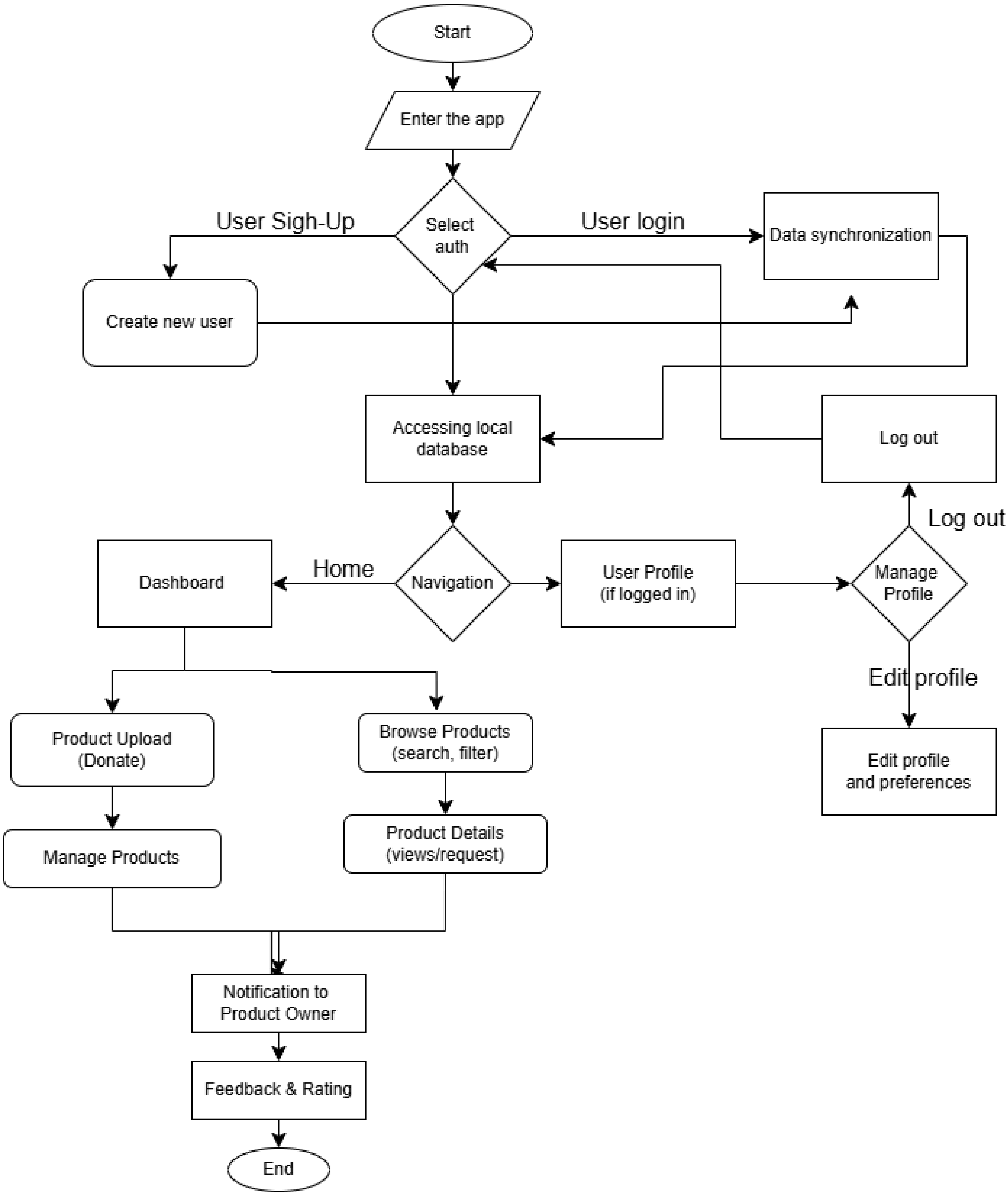

Figure 1: Flow Chart of Reuse Hub Architecture

The above Figure 1 illustrates the high-level user interaction flow of Reuse Hub, showcasing how various components such as item listing, request handling, and user roles (Donor and Seeker) operate within the app. The diagram primarily focuses on the frontend-driven flow, from authentication to donation completion.

7.2. Schema Diagram

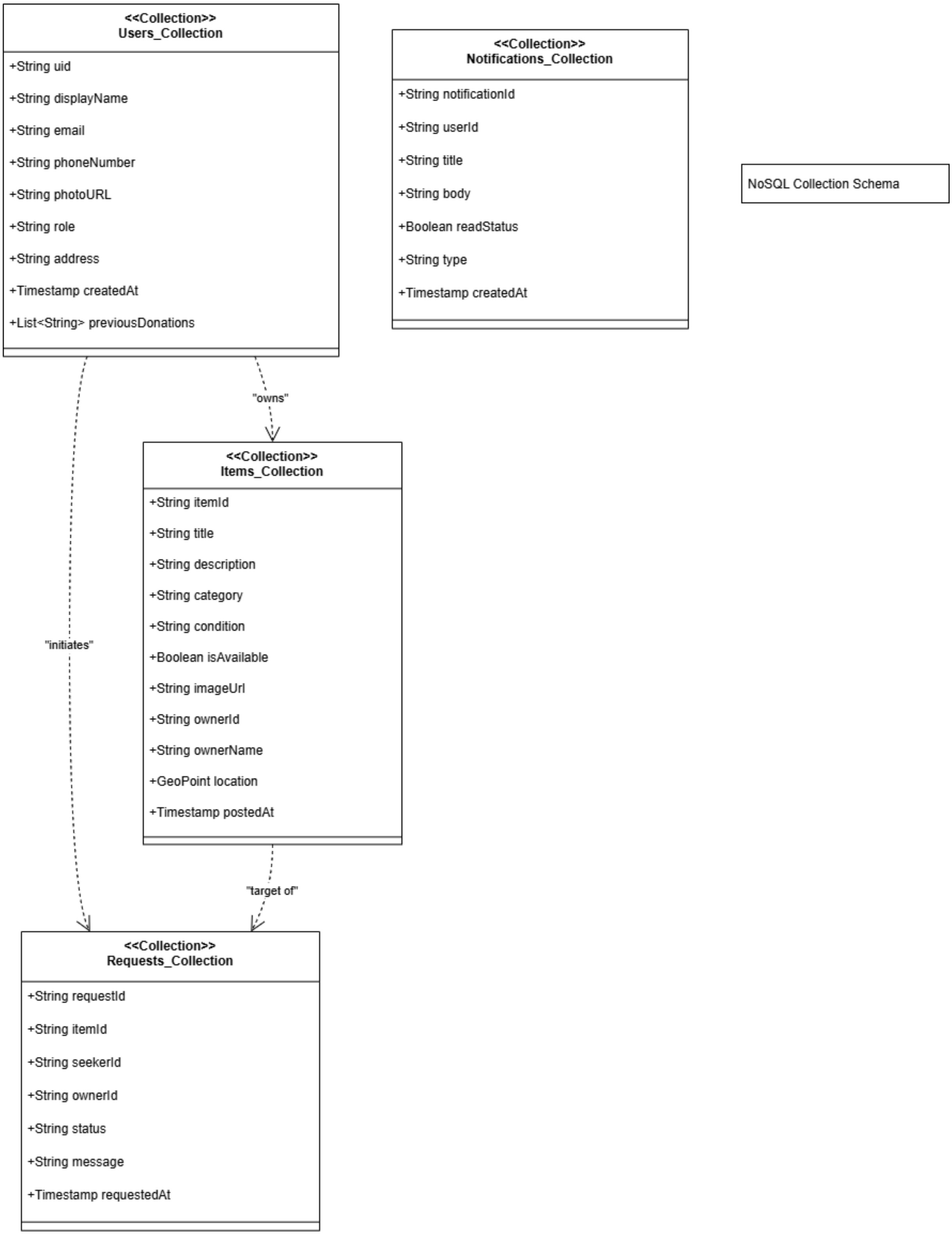

Figure 2: Database Schema Diagram of Reuse Hub

The above Schema Diagram illustrates the database schema for Reuse Hub, showing the tables such as `items` and `users`, their fields, and relationships between them.

7.3. ERD (Entity Relationship Diagram)

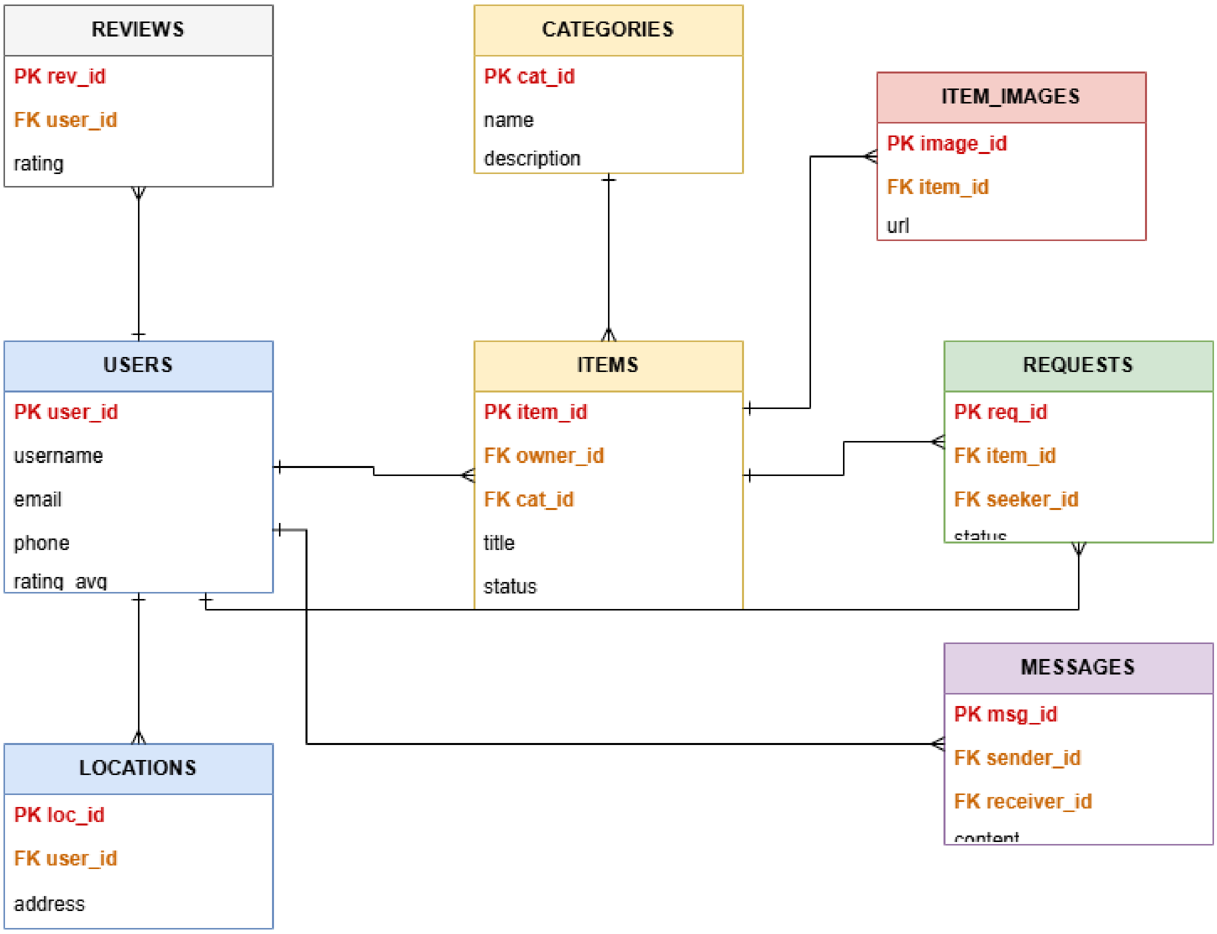

Figure 3: Entity Relationship Diagram of Reuse Hub

The above ERD illustrates the entity-relationship diagram for Reuse Hub, showing the entities, their attributes, and relationships between them. The ERD helps in understanding the data model and how different entities interact with each other.

7.4. Timeline (Gantt Chart)

The base timeline for the development of Reuse Hub is as follows,

Task	W 1-4	W 5-7	W 8-9	W 10-11	W 12	W 14	W 15	W 16
Requirements & UI Wireframing	✓		✓					
Authentication + DB		✓	✓					
Item Listing & Upload Module			✓	✓				
Nearby Item Discovery				✓				
Request & Donation Management			✓	✓	✓			
Chat & Notification System						✓		
UI Polish & Accessibility						✓	✓	
Final Testing & Deployment								✓

Table 1: Development Timeline of Reuse Hub

7.5. UI Mockups

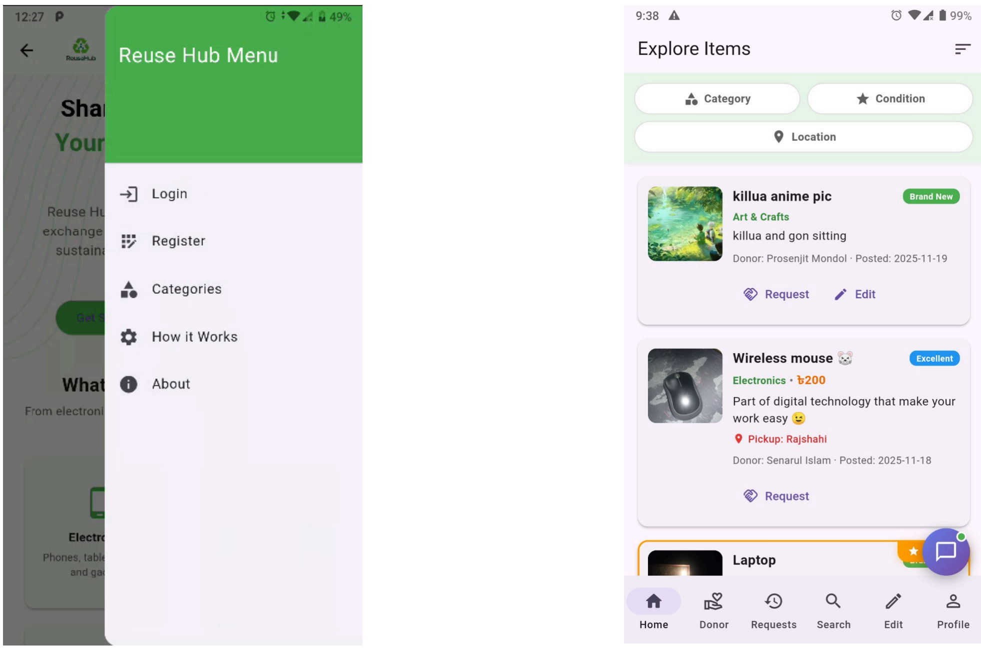

Figure 4: Dashboard and Donation Categories (Food, Cloth, Electronics, Books)

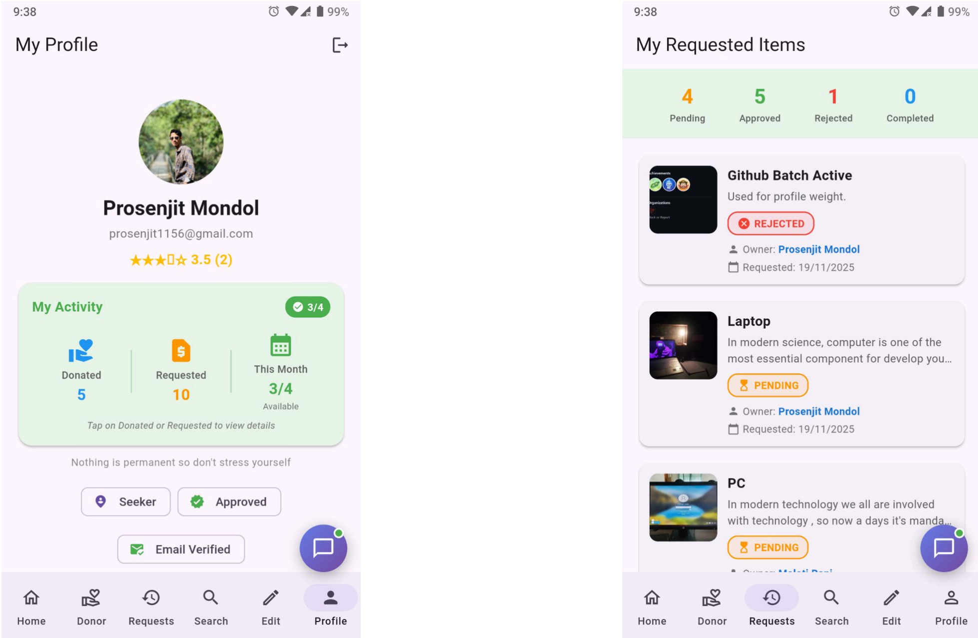

Figure 5: User Profile & Requested Items UI concept

8. Future Plans

1. In future iterations, Reuse Hub can be expanded to support location-based recommendations.
2. Implementing donation history tracking for users.
3. Adding user reviews to build trust between donors and seekers.
4. Implementing item delivery logistics integration.

9. Result

The expected outcome of Reuse Hub is a fully functional, cross-platform application that enables users to donate and collect reusable household items efficiently.

10. References

[1] United Nations, "Sustainable Consumption and Production." Accessed: July 18, 2025. [Online]. Available: https://www.un.org/sustainabledevelopment/sustainable-consumption-production/

[2] OLX, "Online Marketplace Bangladesh." Accessed: July 18, 2025. [Online]. Available: https://www.olx.com/

[3] Flutter, "Flutter Documentation." Accessed: July 18, 2025. [Online]. Available: https://flutter.dev/docs

[4] Firebase, "Firebase Documentation." Accessed: July 18, 2025. [Online]. Available: https://firebase.google.com/docs